

Yan Li

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Research Interests

Trustworthy Medical AI; Multimodal AI for Healthcare; Medical Vision-Language Models; Clinical AI Agents.

Education

Beijing University of Posts and Telecommunications

Sept. 2025–present | Beijing, China

M.S. in Management Science and Engineering

Advisor: Yawen Li; Co-advisor: Guanhua Ye.

University of Electronic Science and Technology of China

Sept. 2021–July 2025 | Chengdu, China

Dual B.S. in Finance and Electronic Information Engineering

Thesis Advisor: Bo Liu.

Research Experience

CAPRA: Multimodal Auditing under Missing Clinical Metadata

ACM MM 2026 Oral

Lead Student Researcher, Beijing University of Posts and Telecommunications

- Led CAPRA, a semantic-visual multimodal framework that fuses medical images with proxy-axis tokens and calibrated posteriors for missing-metadata subgroup auditing.
- Validated fundus, dermoscopy, and chest-radiography tasks under distribution shift, improving worst-group accuracy in 14 of 15 learner-dataset settings.

Selective-Dependence Auditing for Medical LLM Fairness

Under Review

Lead Student Researcher, Beijing University of Posts and Telecommunications

- Led the Selective-Dependence Audit, separating medical judgments, care pathways, support actions, and interface effects to distinguish prohibited dependence from appropriate clinical adaptation.
- Built ClinSDA-400 and evaluated six LLMs across 1,075,020 responses in semantic-label, JSON, and MCQ formats, isolating audit displacement caused by option order and answer-symbol mapping.

MedFuse: Anatomical-Prior Fusion for Retinal Lesion Segmentation

FCS 2026

Lead Student Researcher, Beijing University of Posts and Telecommunications

- Led technical development with an ophthalmologist at Peking University First Hospital, integrating RGB fundus features with retinal vascular priors for diabetic-retinopathy lesion segmentation.
- Evaluated a hierarchical dual-encoder fusion pipeline on DDR and IDRiD across five backbones, including annotation-free priors from a multimodal large language model.

Peer-Reviewed Publications

ACM MM 2026 ORAL Beyond Metadata: CAPRA for Hidden Subgroup Analysis under Missing Metadata in Medical Imaging

Yawen Li, **Yan Li**, Zhe Xue, Yingxia Shao, Meiyu Liang, and Guanhua Ye^{*}.

Proceedings of the 34th ACM International Conference on Multimedia, 2026.

ACM MM 2026 What Does Your Short-Answer VQA Score Actually Measure? Evaluator-Dependent Instability in Multimodal Short-Answer Benchmarks

Guanhua Ye, Jingbin Niu, **Yan Li**, Meiyu Liang, Zhe Xue, Yingxia Shao, and Yawen Li^{*}.

Proceedings of the 34th ACM International Conference on Multimedia, 2026.

AAAI 2026 Rethink Representation Learning for Questionnaire Data

Guanhua Ye, Jifeng He, **Yan Li**, Junping Du, Zhe Xue, Yingxia Shao, Meiyu Liang, and Yawen Li^{*}.

Proceedings of the AAAI Conference on Artificial Intelligence, 40(33):27782–27790, 2026.

TKDE 2025 **FeBT: A Feature Balancing Transformer for Corporate ESG Forecasting**
Yawen Li, Mengyu Zhuang, Guanhua Ye^{*}, **Yan Li**, Junheng Wang, Jinyi Zhou, and Pengfei Zhang.
IEEE Transactions on Knowledge and Data Engineering, 2025.

FCS 2026 **MedFuse: A Multi-source Data Fusion Framework for Diabetic Retinopathy Lesion Segmentation**
Jiantong Du, **Yan Li**, Yawen Li, Liwen Liao, Zhihao Zhao, and Guanhua Ye^{*}.
Frontiers of Computer Science, 20(3):2003625, 2026.

Conference Presentations

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| “Trust-free Personalized Decentralized Learning”
<i>Oral Presentation at IEEE BigComp 2026, Guangzhou, China.</i> | Feb. 5, 2026 |
| “Slot-Based Evidence Routing for Budgeted Multi-Agent Coordination”
<i>Oral Presentation at CINT 2026, Hangzhou, China.</i> | Mar. 16, 2026 |
| “Auditing Historical Clinical Queries with Missing Availability Time”
<i>Oral Presentation at CCF BigData 2026, Harbin, China.</i> | Aug. 21, 2026 |

Teaching

Beijing University of Posts and Telecommunications

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| Big Data and Intelligent Management
<i>Teaching Assistant, graduate course, School of Economics and Management</i> | Fall 2025 |
| Operating Systems
<i>Teaching Assistant, undergraduate course, School of Computer Science</i> | Spring 2026 |

Honors and Awards

University of Electronic Science and Technology of China

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| Certificate for Honors Research | 2025 |
| “Academic Star” Award of School of Management and Economics (sole recipient) | 2024 |
| Outstanding Student Scholarship | 2024 |
| Model Student Scholarship | 2023 |